

Comprehensive Project Status, Delta Analysis & Master Sprint Roadmap

v5.0 NodeTool-First Architecture · Audited Codebase & Net Effort Plan · August 2026

1. Executive Summary & Project Context

This master implementation plan establishes an audited, high-precision roadmap tailored to the client's v5.0 NodeTool-First specification. Crucially, **time is not allocated to tasks already completed in the codebase** (such as scrapers, standard image resizing, Dutch AI descriptions, basic MCP tools, and recommendation algorithms). Instead, this document details the exact **Architectural Delta** (what exists vs. what changes) and allocates timeline solely to the new NodeTool integration, Editor Seam contract, owned storage, video assembly, and 3D pipelines.

Strict Delivery & QA Protocol:

- **1. Spec Alignment:** Define explicit input/output criteria before building each sprint goal.
- **2. Client Approval:** Client sign-off on sprint specifications.
- **3. Internal QA:** End-to-end verification across Baserow, n8n, fal.ai, and Shopify staging.
- **4. Client Demo Walkthrough:** Interactive inspection grids, testing links, and video walkthroughs.
- **5. Go-Live Verification:** Production cutover with automatic safety locks.

2. Detailed Delta Analysis: Existing Codebase vs. v5.0 Architectural Changes

The following matrix maps every project capability, its current verified completion in the codebase, and the precise architectural modification required by the v5.0 specification:

System Capability	Current Status in Codebase	Required Architectural Change (Delta) in v5.0 Spec	Net Effort Remaining
Scraper Infrastructure	100% COMPLETE (28/28 Brands) All 9 priority & 19 standard brand scrapers built in scrapers/router.py.	Zero Scraper Changes Needed: Scraper router extracts full product specs, dimensions, and Dutch translations directly to Baserow.	0 DAYS (COMPLETED)
Hero BG Removal & Resizing	100% FUNCTIONAL Built in fal_image_processor.py & shopify_pixelbin_remove_bg.py (1000x880 transparent, 1760x1100 white).	Rewire into Intake Chain: Move from standalone CLI execution into n8n automated intake chain calling NodeTool headless API. Save to owned S3 (asset__v1) instead of temporary URLs.	1 Day (Rewiring)
AI Dutch Descriptions	100% COMPLETE Operational in description_ai.py (OpenRouter GPT-4o / Claude).	Token Referencing: Hook into Baserow Prompt_Scene_Templates and Video_Templates to inject descriptions as token parameters ({title}, {description}).	0.5 Day (Integration)
Shopify Storefront Sync	100% FUNCTIONAL Operational in baserow_shopify_sync.py for Woonbloq (1) and Binnen (3).	Field Mapping & Safety Lock: Map new S3 URL fields (clean_hero_url, lifestyle_chosen_url, video_urls, 3D_urls) and enforce enrichment_status = 'edited' safety lock.	0.5 Day (Schema update)
Agentic MCP Suite	100% FUNCTIONAL BASE Native Python MCP server in mcp_server.py (Baserow & Shopify tools).	Add Jobs Tool & Tri-MCP: Add create_enrichment_job and get_job_status tools to mcp_server.py; configure Claude Cowork to connect Baserow MCP + NodeTool MCP + FableCut MCP.	1.5 Days (Tri-MCP config)
Recommendation Engine	100% ALGORITHMS READY Pre-built in recommendation_engine.py (3-tier vector cosine similarity + Quality score + FBT bundle engine).	Staging Sync & Metafields: Execute sync_recommendations_to_shopify.py against live store metafields and connect FBT pairs into the bundle image layout engine.	1 Day (Staging Sync)

NodeTool Creative Core	NEW IN v5.0 SPEC Not yet deployed.	Deploy & Validate (\$6): Self-hosted NodeTool Docker deployment with BYOK fal.ai key; execute Phase 0 6-gate playground validation; setup headless execution API.	2.5 Days (Phase 0 setup)
The Editor Seam Service	NEW IN v5.0 SPEC Not yet deployed.	Build Thin Seam Service: FastAPI router resolving Baserow formula links into NodeTool/FableCut/3D Viewer; return watched folder ingester → S3 versioned upload → Baserow edited=true lock.	2.5 Days (Seam Build)
Video Production Pipeline	NEW IN v5.0 SPEC Basic video prototypes only.	Dual-Class Video Architecture: Remotion/FFmpeg programmatic assembly (zero baked text) + Seedance 2.0 timeline-prompt lifestyle ads with @product image anchors + FableCut agent bench.	3.5 Days (Video Build)
3D Pipeline & Viewer	NEW IN v5.0 SPEC Not yet deployed.	Hunyuan + glTF/USDZ: fal Hunyuan 3D connector + n8n glTF-Transform compression + USDZ conversion + lightweight <model-viewer> review page.	2.5 Days (3D Build)

3. Streamlined Master Timeline & Sprint Roadmap (Net Effort)

By leveraging existing codebase foundations, the project timeline is condensed to **real active engineering days**, eliminating redundant milestones:

Sprint / Phase	Core Focus & Architectural Deliverables	Active Duration	Dependencies / Key Inputs
Phase 0 (Mandatory)	NodeTool Playground Validation (\$6): Standalone Docker testing of Canvas, Sketch Masking on furniture, Video Timeline, Seedance 2.0, Headless CLI & MCP.	2.5 Working Days	Docker host, fal.ai key, 10 test furniture products.
Phase 1	Storage, Baserow Schemas & n8n System Glue: S3/MinIO bucket setup, Baserow Jobs & Template tables, n8n intake (A) & grid marker (B) triggers.	2 Working Days	Baserow admin token, S3 credentials, n8n instance.
Phase 2	The Editor Seam & Rewired Image Pipelines: Deploy Seam microservice, rewire existing resizing into intake chain, on-demand lifestyle generation & correction loop.	2.5 Working Days	NodeTool API endpoints, Baserow formula links.
Phase 3	Claude Cowork Tri-MCP & FableCut Setup: Extend Python MCP server for Jobs queue, deploy FableCut Docker, test Cowork JSON timeline patching.	2 Working Days	Claude Cowork / Desktop setup, Anthropic key.
Phase 4	Automated Video Production Pipeline: Remotion/FFmpeg assembly (zero baked text), Seedance lifestyle ads with @product anchors, Shopify media push.	3.5 Working Days	Seedance 2.0 / FLUX 3 via fal.ai, Remotion templates.
Phase 5	3D Pipeline Integration & <model-viewer>: fal image-to-3D, glTF compression, USDZ conversion, review page & Shopify 3D metafield sync.	2.5 Working Days	fal.ai 3D endpoint, glTF-Transform, USDZ tools.
Phase 6	Recommendation Engine & FBT Live Sync: Push pre-built 3-tier matching and Frequently Bought Together bundles to live Shopify storefront.	1 Working Day	Existing recommendation_engine.py, Shopify theme.
Phase 7	Composite Bundle Image Generation: Multi-product layout canvas, edge-blending & contact shadows via NodeTool/fal inpainting.	2.5 Working Days	FBT bundle pairs, NodeTool inpaint workflows.
Phase 8	Metadata, Options & Dynamic QR Codes: Dynamic vector QR codes with UTM tracking, configurable product variation matrix.	2 Working Days	Shopify product handles & QR asset storage.
Phase 9	AI Shopping Assistant Chatbot & Go-Live: Catalog vectorization, chatbot widget embedding on frontend, end-to-end QA & production cutover.	3.5 Working Days	Frontend store access, embedding API key.

4. Streamlined Daily Sprint Execution Breakdown

Sprints focus purely on new delta deliverables and integration checkpoints:

Phase 0: NodeTool Playground Sandbox Validation (Duration: 2.5 Working Days — Mandatory per \$6)

- **Day 1:** Deploy self-hosted NodeTool Docker container with fal.ai key. Load 10 sample products manually.
- **Day 2:** Validate Canvas lifestyle workflows & Layered Sketch Editor (masking/inpaint on furniture).
- **Day 3 (Deliverable 0 Checkpoint - Half Day):** Validate Video Timeline, Seedance 2.0 timeline generation, Headless CLI execution, and NodeTool MCP. Present verification matrix to client. (If partial fail, activate pre-decided fallback).

Phase 1: Storage, Baserow Schemas & n8n System Glue (Duration: 2 Working Days)

- **Day 1:** Provision S3/MinIO bucket (binnen-catalog-assets) with versioning (asset__v{n}). Create Baserow Jobs, Prompt_Scene_Templates, and Video_Templates tables.
- **Day 2 (Deliverable 1 Checkpoint):** Configure n8n webhook listeners for Intake Trigger (A) and Multi-Row Marker (B). Deploy n8n Job Queue runner with live Baserow progress write-back.

Phase 2: The Editor Seam & Rewired Image Pipelines (Duration: 2.5 Working Days)

- **Day 1:** Build The Editor Seam microservice (FastAPI router resolving Baserow formula links into NodeTool/FableCut + watched folder export listener).
- **Day 2 (Deliverable 2 Checkpoint):** Rewire existing fal_image_processor.py resizing logic into n8n intake chain calling NodeTool headless API. Push versioned assets to S3 and link to Baserow.
- **Day 3 (Deliverable 3 Checkpoint - Half Day):** Connect On-Demand Generative Tier (2-3 lifestyle variants per product from Baserow prompt templates) and test 3-tier Image Correction Loop.

Phase 3: Conversational Claude Cowork & Tri-MCP Suite (Duration: 2 Working Days)

- **Day 1:** Add create_enrichment_job and get_job_status tools to existing mcp_server.py. Deploy FableCut Docker container.
- **Day 2 (Deliverable 4 & 5 Checkpoints):** Connect Baserow MCP + NodeTool MCP + FableCut MCP for Claude Cowork (Entry Point C). Test conflict-safe JSON timeline patching and deliver operational demo.

Phase 4: Automated Video Production Pipeline (Duration: 3.5 Working Days)

- **Day 1:** Build Remotion / FFmpeg programmatic rendering templates for Functional Videos (stills + specs + price overlays; zero baked text).
- **Day 2:** Author timeline-prompt recipes in Video Templates table using Seedance 2.0 with @product anchors.
- **Day 3:** Configure n8n video assembly pipeline with dynamic audio syncing. Wire FableCut & NodeTool into the Seam.
- **Day 4 (Deliverable 6 & 7 Checkpoints - Half Day):** Push generated product videos directly to Shopify media galleries on staging.

Phase 5: 3D Asset Pipeline Integration & (Duration: 2.5 Working Days)

- **Day 1:** Connect fal.ai Hunyuan image-to-3D connectors with parameter escalation ladders on reject.
- **Day 2:** Build automated n8n post-processing: glTF-Transform compression + USDZ conversion for iOS AR.
- **Day 3 (Deliverable 8 Checkpoint - Half Day):** Host lightweight <model-viewer> review page. Deploy 3D models to Baserow and Shopify.

Phase 6: Recommendation Engine & FBT Staging Sync (Duration: 1 Working Day)

- **Day 1 (Deliverable 9 Checkpoint):** Run pre-built sync_recommendations_to_shopify.py to sync 3-tier recommendations and Frequently Bought Together (FBT) bundles to Shopify metafields. Showcase live PDPs.

Phase 7: Composite Multi-Product Bundle Engine (Duration: 2.5 Working Days)

- **Day 1:** Build canvas layout engine to arrange FBT pairs (e.g. Sofa + Coffee Table + Rug) onto a unified room canvas.
- **Day 2-3 (Deliverable 10 Checkpoint):** Program contact shadows and edge-blending filters via NodeTool/fal inpaint. Sync composite graphics to Shopify collections.

Phase 8: Metadata, Configurable Products & QR Engine (Duration: 2 Working Days)

- **Day 1:** Build dynamic QR code generator assigning uniquely scannable codes with UTM tracking to incoming products.
- **Day 2 (Deliverable 11 Checkpoint):** Configure backend variation logic separating configurable variants from standalone products and sync QR assets to Shopify.

Phase 9: AI Shopping Assistant Chatbot & Final Go-Live (Duration: 3.5 Working Days)

- **Day 1:** Vectorize complete product catalog (embeddings for titles, Dutch descriptions, 13 style vectors, prices, stock).
- **Day 2:** Embed and style AI Shopping Assistant chatbot widget onto the storefront frontend.
- **Day 3 (Deliverable 12 Checkpoint):** Test and refine dynamic catalog search filters through chatbot queries.
- **Day 4 (Deliverable 13 Checkpoint / Go-Live - Half Day):** Full-scale end-to-end QA across all user paths, Baserow, n8n, S3, NodeTool, and Shopify webshops. Live cutover!

5. Pre-Decided Fallback Matrix (Zero-Delay Risk Mitigation)

To guarantee that technical blockers never stall delivery, pre-decided fallbacks slot in behind the exact same Editor Seam contract:

Component	Primary Solution	Fallback Trigger Criteria	Pre-Decided Fallback Solution
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Image Masking	NodeTool Sketch Editor	Masking precision inadequate or laggy in Phase 0.	InvokeAI on Vast.ai on-demand GPU (~\$0.25/hr booted by n8n) or Gemini image API.
Human Video Timeline	NodeTool Video Timeline	Timeline trims/splits unstable in Phase 0.	FableCut project wrapper for human timeline too (swapping active project JSON).
Workflow Execution	NodeTool Headless API/CLI	Headless node execution flaky.	n8n directly calling fal.ai APIs with prompt templates stored in Baserow.
Agentic Tools	NodeTool MCP	NodeTool MCP immature.	Baserow MCP + FableCut MCP only; NodeTool accessed via the Seam.

Next Engineering Action: Proceed with Phase 0 Task 0.1 (NodeTool Docker sandbox setup and 6-gate validation).